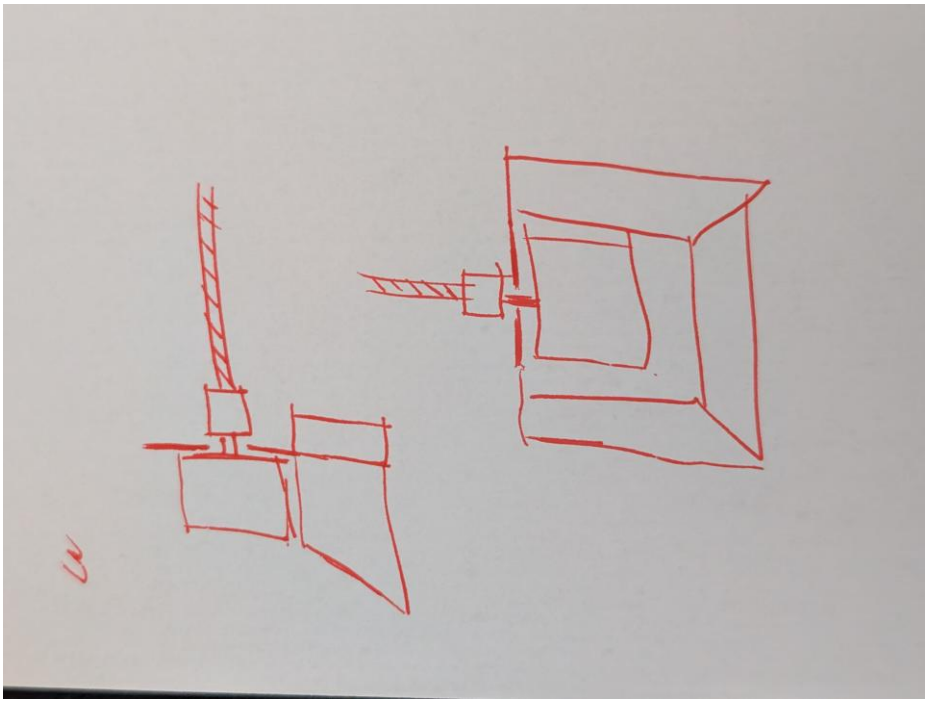
BUILD MANUAL

Automated Macro Focus-Stacking Rail

MVP printing, mechanical assembly, alignment, and acceptance



Design concept: motor-end support, centered screw, guide rods, and broad mounting feet.

Version 1 | 150 mm verified travel | Anet A8-derived hardware

1. Read This First

This guide covers the mechanical MVP only: printing the seven custom parts, assembling the two-rod stage, aligning the carriage and lead screw, and confirming safe motion.

Core principle: The rods guide. The screw pushes. The base supplies stiffness. Tighten the screw nut only after the carriage moves freely.

Build Envelope

Item	MVP value
Guide rods	Ø8 × 380 mm, 80 mm centers
Lead screw	TR8×8(P2), 345 mm, 8 mm/rev
Axis height	30 mm above base datum
Bearing spacing	80 mm front-to-rear
Verified travel	150 mm
Sample platform	80 × 80 mm

Safety and Handling

- ☐ Disconnect power while assembling, aligning, or tightening the coupler.
- ☐ Do not use motor screws long enough to contact internal windings.
- ☐ Deburr printed holes and metal rod ends before sliding bearings into place.
- ☐ Never force a binding carriage. Find and correct the alignment error.
- ☐ Keep fingers clear of the coupler and screw during powered tests.

Acceptance Standard

The MVP is ready for powered testing when the carriage traverses smoothly by hand, the screw does not steer it sideways, the endstop triggers before a hard collision, and at least 15 mm remains at the configured far limit.

2. Parts and Hardware

Printed Parts

STL	Part	Nominal envelope
01	Motor-end block	75 × 140 × 65 mm
02	Far-end block	45 × 140 × 55 mm
03	Carriage	110 × 134 × 6 mm
04	Floating nut carrier	30 × 40 × 23 mm
05	Sample platform	80 × 80 × 5 mm
06	Endstop bracket	55 × 32 × 34 mm
07	Endstop trigger	40 × 15 × 14 mm

Reused Motion Hardware

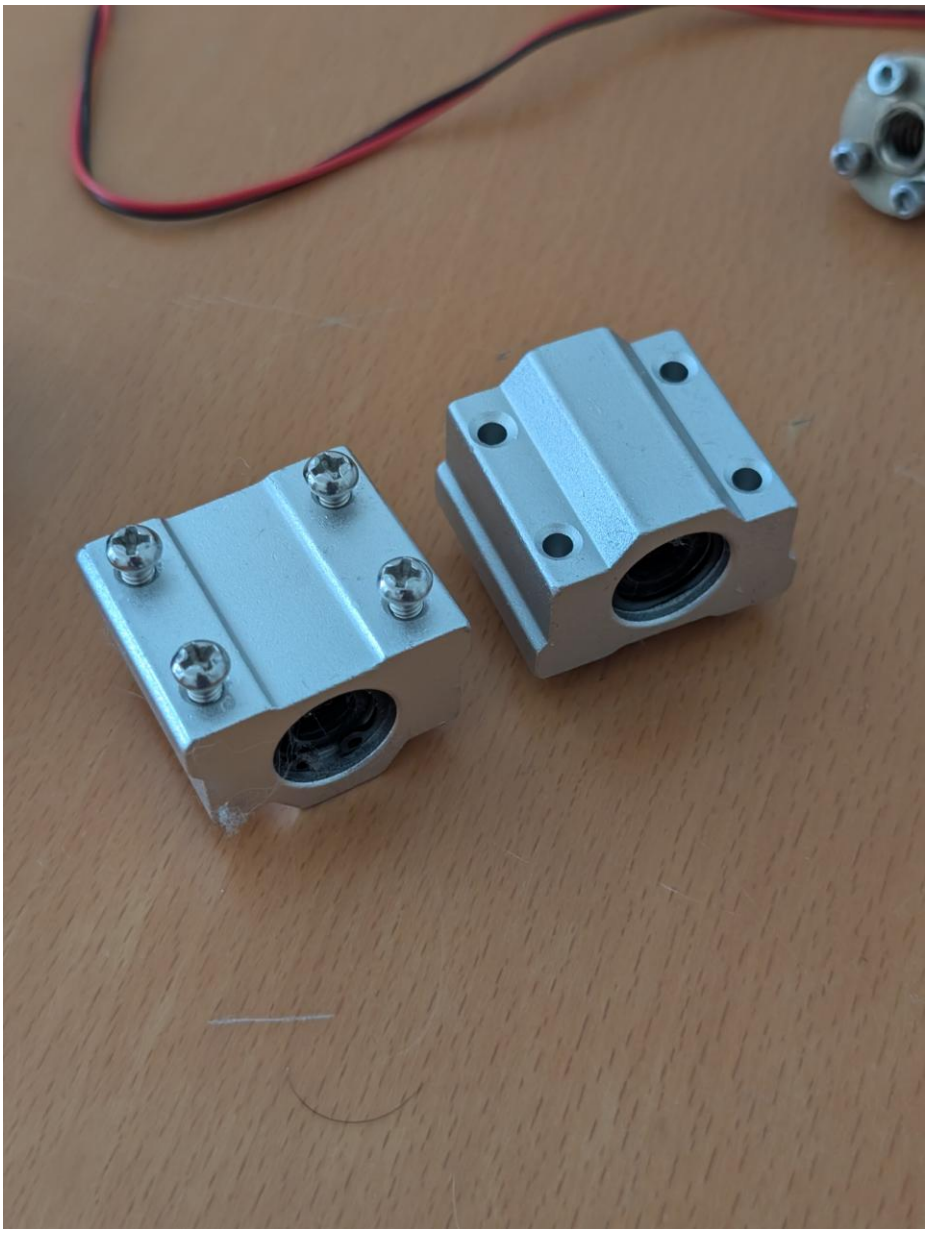
- ☐ 2 × Ø8 mm smooth rods, approximately 380 mm long
- ☐ 4 × SC8UU linear-bearing blocks
- ☐ 1 × TR8×8(P2) lead screw, approximately 345 mm long
- ☐ 1 × 22 mm flanged T8 brass nut
- ☐ 1 × NEMA 17 motor, model 42SHDC3025-24B or equivalent
- ☐ 1 × Ø18.6 mm flexible helical coupler
- ☐ 1 × roller-lever microswitch/endstop

Provisional Fasteners

Interface	Hardware
Motor to block	4 × M3×14–16; verify engagement
SC8UU blocks	Up to 16 × M4×10–12
T8 nut	Reuse four existing M3 screws
Nut carrier	4 × M4×16 + washers/nuts
Platform	4 × M4×16 + washers/nuts
Trigger	2 × M3×16 + washers/nuts
Rod retention	4 × M4; tap/form printed pilots
End blocks to base	M5 + large washers

3. Identify the Reused Parts

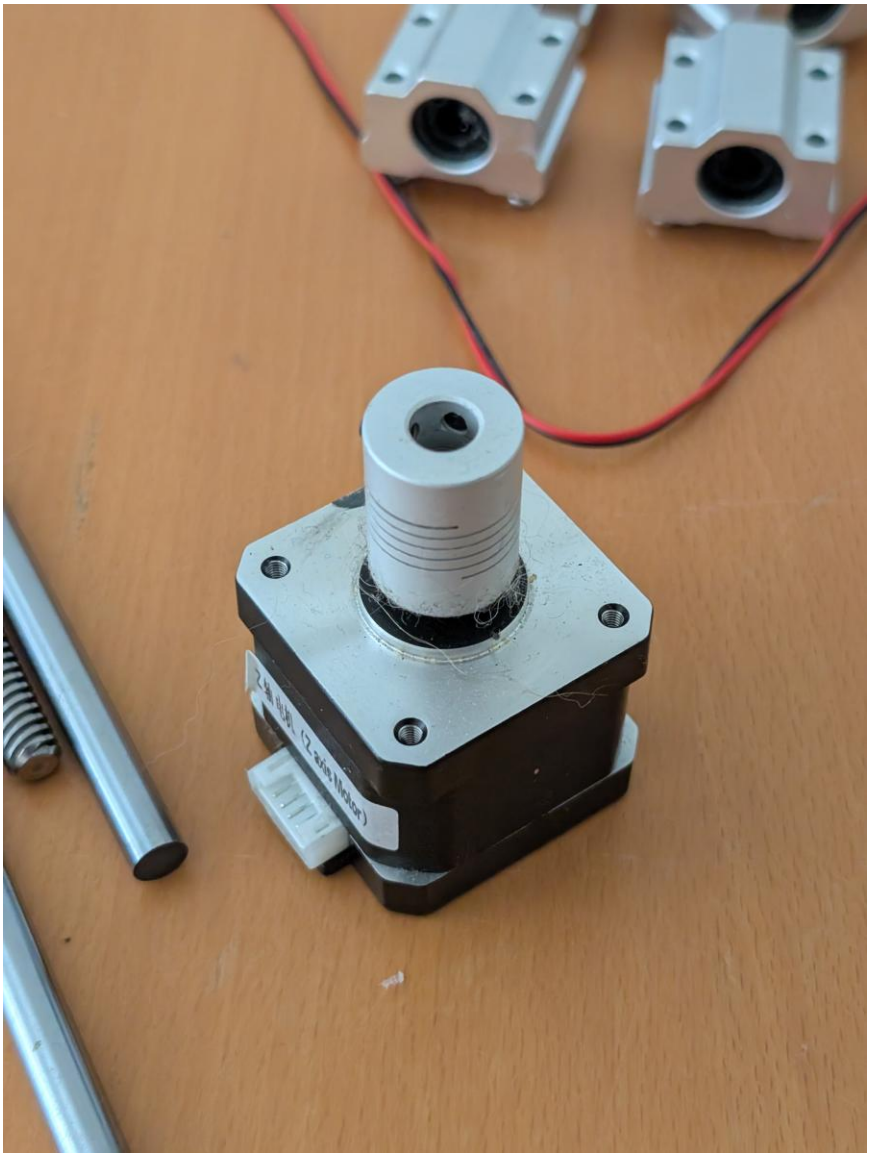
SC8UU Bearing Blocks



SC8UU bearing blocks. Use all four: two blocks on each guide rod.

Each block is 30 mm along travel, 34 mm across the rail, and 22 mm high. The shaft center is 11 mm below its mounting face. Mounting threads are M4.

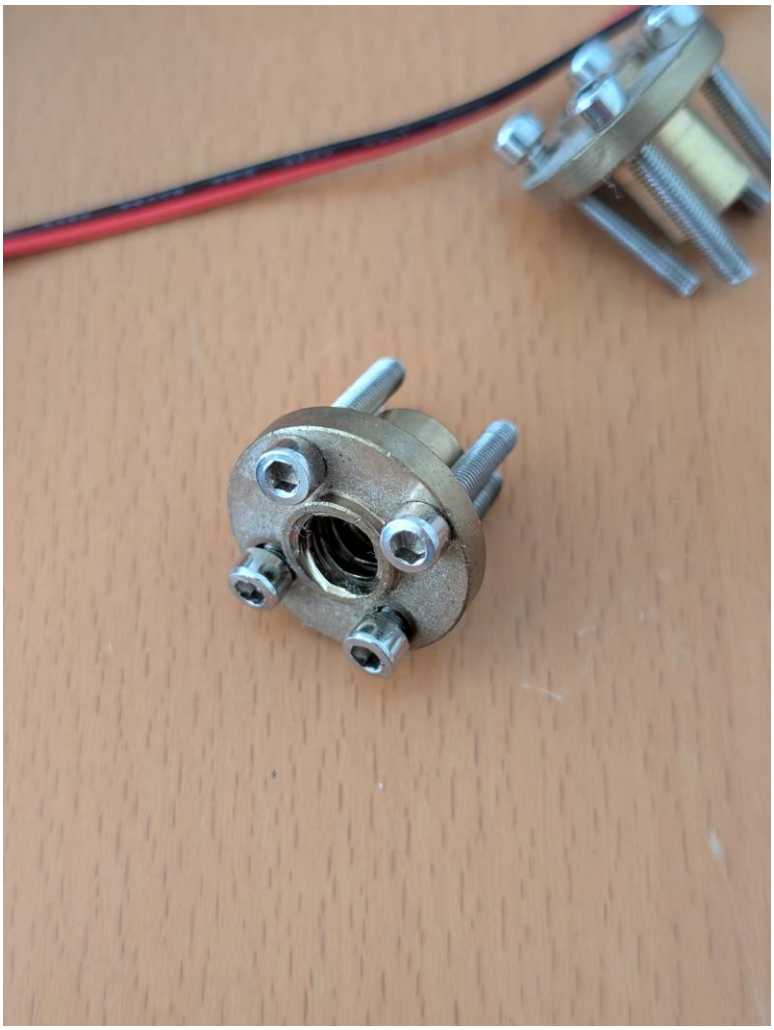
Motor and Coupler



NEMA 17 motor with the Ø18.6 mm helical coupler already installed.

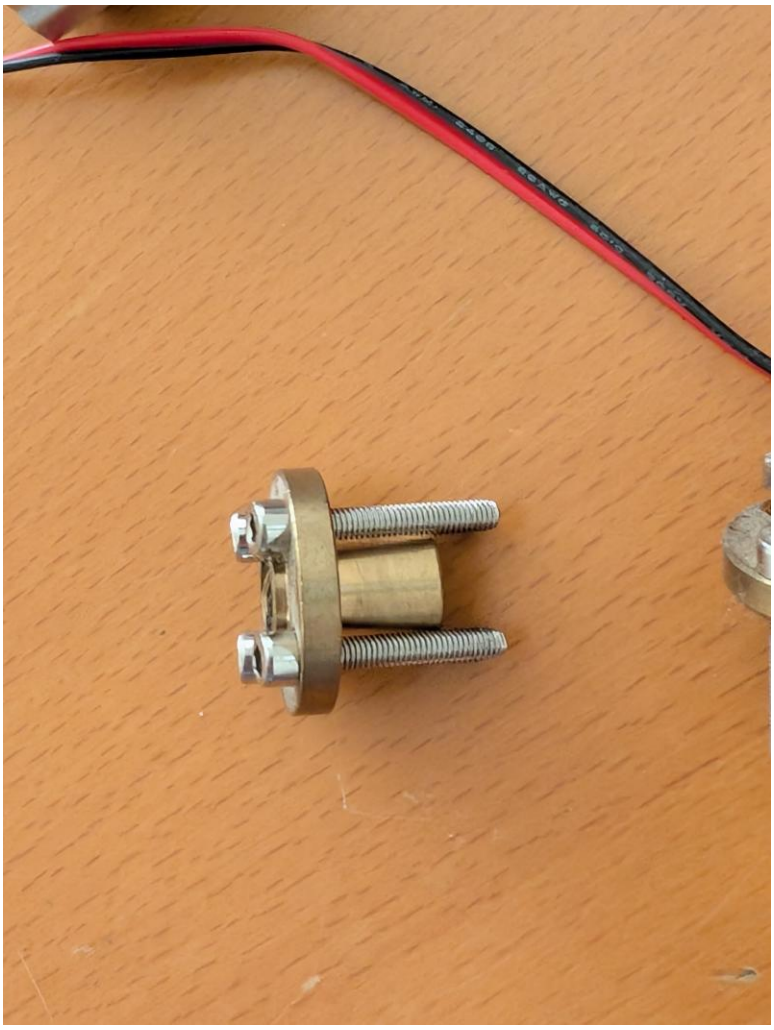
The motor mounts to the vertical motor-end plate using the four front M3 holes. The coupler passes through the central service opening; keep both set screws accessible.

4. T8 Nut and Endstop



Front view of the one-piece flanged T8 nut and its four mounting screws.

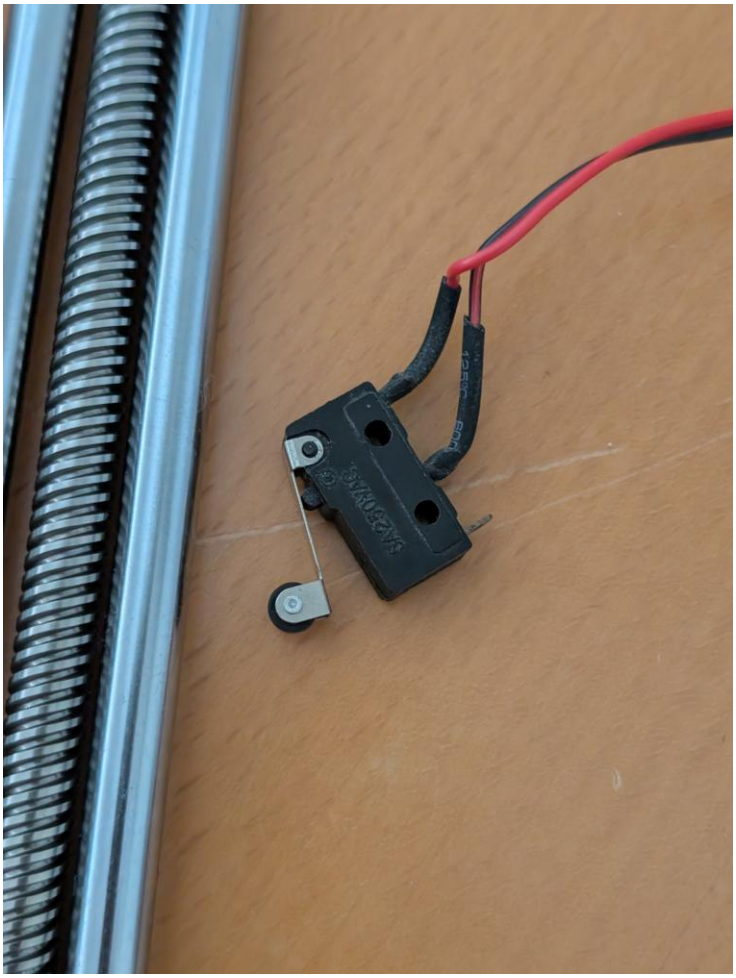
The nut flange is 22 mm diameter, 15 mm overall height, and 3.5 mm thick. Its four M3 clearance holes form an approximately 11 mm square.



Side view of the brass nut. The printed carrier provides lateral alignment compliance.

Do not rigidly locate the nut first: Leave the floating carrier loose until the carriage, rods, screw, and coupler establish the natural screw axis.

Roller-Lever Endstop



Roller-lever microswitch. Mount it so the trigger approaches the roller gradually.

5. Printing and Preparation

Slicer Setup

- ☐ PLA prototype; upgrade material only after the fit is proven.
- ☐ 0.4 mm nozzle and 0.2 mm layer height.
- ☐ At least 4 walls/perimeters and 5 top/bottom layers.
- ☐ 40–60% infill for end blocks, carriage, and nut carrier.
- ☐ 30–50% infill for the platform, endstop bracket, and trigger.
- ☐ Use drop-to-bed and center/arrange: the STL files retain assembly coordinates.

Recommended Orientation

Part	Orientation
Motor/far blocks	Broad mounting foot on bed
Carriage/platform	Largest flat face on bed
Endstop bracket	Base foot on bed
Trigger	Broad attachment pad on bed
Nut carrier	Upside down, broad 30 × 40 mm flange on bed

Post-Print Preparation

1. **Deburr.** Remove elephant's foot, strings, and support residue from every mating face.
2. **Check holes.** Test M3/M4/M5 clearance holes without forcing screws.
3. **Prepare pilots.** Tap or thread-form the 3.3 mm M4 rod-retention and trigger holes.
4. **Test rods.** Slide both rods through both end blocks before installing the carriage.
5. **Test bearings.** Run every bearing block on both rods and reject rough or loose combinations.

6. Build the Carriage

Install the Four Bearing Blocks

6. **Orient the carriage.** Place its flat sample-platform face upward and the bearing mounting face downward.
7. **Place the blocks.** Use two SC8UU blocks per rod location, with 80 mm between front and rear centers.
8. **Start all M4 screws.** Engage every screw loosely before tightening any one block.
9. **Tighten progressively.** Alternate diagonally so the blocks remain flat against the carriage.

Check: The four 8 mm bores should form two visually straight, parallel lines. A twisted block will create immediate binding.

Install the Floating Nut Carrier

10. **Mount the brass nut.** Attach its flange to the vertical carrier face using the four M3 screws.
11. **Attach the carrier.** Use four M4 bolts through the carriage adjustment slots; leave them loose.
12. **Verify clearance.** The screw should pass through the carrier without touching printed material.

Install the Flat Platform

13. **Align the pattern.** Match the platform's 60 × 20 mm hole pattern to the carriage.
14. **Bolt it down.** Use M4 hardware and washers; tighten until rigid without bowing either plate.

7. Assemble the Rod Stage

15. **Secure the motor-end block.** Place it on the future base datum or a known-flat temporary surface.
16. **Insert both rods.** Feed the two rods through the motor-end passages. Do not fully tighten the M4 retainers.
17. **Slide on the carriage.** Support its weight and pass both rods through all four bearing blocks.
18. **Add the far-end block.** Slide it onto both rod ends and leave its base slots loose.
19. **Align by feel.** Move the carriage repeatedly and adjust the far end until resistance is even and minimal.
20. **Lock the rods.** Tighten the motor-end retainers first, then the far-end retainers gradually.

Critical gate: With the lead screw still absent, the carriage must move smoothly across the required travel. If it binds now, do not install the screw.

Diagnosing Rod-Side Binding

Symptom	Likely cause	Correction
Tight everywhere	Rod spacing mismatch	Loosen far block and bearing screws
Tight near one end	Rods not parallel	Realign far block
Repeating roughness	Dirty/damaged rod or bearing	Clean, rotate, or replace
Carriage rocks	Loose blocks or worn bearing	Retighten or replace

8. Install the Motor and Lead Screw

21. **Mount the motor.** Fasten the NEMA 17 to the outer face of the motor-end plate with four M3 screws.
22. **Check the coupler.** Confirm its motor-side set screw is tight and accessible through the service opening.
23. **Insert the lead screw.** Push approximately 6.6 mm into the coupler without bottoming both shafts together.
24. **Tighten the screw side.** Secure the coupler set screw against the lead screw.
25. **Engage the nut.** Rotate the screw by hand through the brass nut; never force the thread.
26. **Find natural alignment.** Run the carriage through its travel while the carrier-to-carriage bolts remain loose.
27. **Lock the carrier.** Tighten the carrier evenly where the screw rotates with the least side load.

Expected screw: TR8×8(P2): 2 mm thread pitch, four starts, and 8 mm of nut travel per full revolution.

Screw-Side Acceptance

- ☐ The screw turns by hand through the full travel.
- ☐ The carriage does not visibly move sideways as the screw rotates.
- ☐ The coupler remains centered and does not whip.
- ☐ The far screw end remains floating and clear of the far block.

9. Install and Adjust the Endstop

28. **Mount the bracket.** Place it outside the left guide rod using its slotted base.
29. **Attach the switch.** Orient the roller toward the carriage trigger and leave coarse adjustment available.
30. **Attach the trigger.** Bolt the printed trigger to the reinforced carriage ear.
31. **Install the adjustment screw.** Thread an M4 screw into the trigger's horizontal 3.3 mm pilot.
32. **Move by hand.** Approach the motor end slowly and listen/feel for the switch click.
33. **Set overtravel margin.** The switch must trigger before any bearing, carrier, or carriage feature reaches the motor block.
34. **Lock the adjustment.** Use a jam nut or thread locker only after reliable operation is confirmed.

Verified design position: The modeled home center is 120 mm. At that position the trigger contacts only the roller; no other moving component intersects the fixed structure.

Mount to the Heavy Base

35. **Fix the motor end first.** Use M5 hardware and broad washers through the slotted foot.
36. **Leave the far end loose.** Cycle the carriage several times and let the rods establish alignment.
37. **Tighten the far end last.** Alternate bolts while confirming that motion does not become tighter.

10. Final Mechanical Acceptance

- ☐ Both rods are clean, straight, and retained securely.
- ☐ All four bearing blocks sit flat and remain tight.
- ☐ Carriage moves smoothly with the screw disconnected.
- ☐ Lead screw rotates without steering the carriage.
- ☐ Coupler set screws are tight and accessible.
- ☐ Nut carrier is tight but not side-loading the screw.
- ☐ Endstop activates before hard contact.
- ☐ Far-end software position preserves at least 15 mm clearance.
- ☐ Sample platform is rigid and does not rock.
- ☐ End blocks are securely mounted to a heavy, flat base.

Troubleshooting

Problem	Check first	Action
Binds without screw	Rod parallelism	Loosen far block; realign rods
Binds only with screw	Nut carrier	Loosen carrier and re-center
Periodic tight spot	Bent screw/coupler	Inspect runout and insertion
Carriage rocks	Bearing fasteners	Retighten diagonally
Endstop misses	Roller/trigger alignment	Use both bracket and screw adjustment
Motor vibrates	Speed/acceleration	Reduce motion settings after mechanical check

First Powered Test

After the mechanical checklist passes, home at very low speed and command only short moves. Keep one hand near power. Confirm direction, endstop behavior, settling, and repeatability before attaching a valuable sample.

Appendix A. Design Record

The MVP geometry was checked in Fusion 360 using simplified reference hardware. All seven printable bodies were valid, with zero unintended static intersections. The moving group was checked from a 120 mm home center to a 270 mm far center, producing 150 mm verified travel.

Verification	Result
Printable components	7
Reference groups	5
Static unintended intersections	0
Home unexpected intersections	0
Far-limit intersections	0
Minimum far clearance	15 mm
Mesh degeneracy	0 triangles
Mesh non-manifold edges	0

Source of Truth

Detailed measurements, assumptions, CAD notes, STL verification, and revision history are maintained in agents.md in the MacroScanner4 workspace. Physical parts remain authoritative over nominal internet dimensions.

MVP limitation: This guide covers the first printable mechanical version. Production material, final base selection, cable routing, firmware, and specialized sample holders remain later refinements.